

Honours Thesis - Simulation of Squeezed Light Generation in an Optical Resonator

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I. TODO

- Check if Tong notes are a good reference with MacRae
- Derive three-level semi-classical Hamiltonian
- Write section on interferometric cavities (Fabry-Perot, spherical, etc.)
- Implement and write up Jaynes-Cummings Model

II. PROJECT MOTIVATION

MacRae's Outline:

Quantum fluctuations of the electromagnetic field impose a fundamental limit on measurement sensitivity, known as the standard quantum limit (SQL). Exotic quantum optical states known as "squeezed light" can surpass the SQL. Squeezed states are a hot topic in research, particularly in quantum communication, metrology, and optical quantum computing. This project will explore theoretical models predicting strong squeezing when atoms are placed inside an optical cavity. The student will develop and run numerical simulations to investigate the conditions for optimal squeezing, using tools from quantum optics and cavity QED theory. Through this work, the student will gain experience in programming, numerical methods, and the theoretical framework of quantum light-matter interactions.

The past hundred years since the formulation of quantum mechanics have led to unrivaled technological progress, especially in recent years. With greater understanding of and ability to control quantum phenomena further possibilities are becoming available. These include quantum key distribution (QKD), quantum communication, and quantum computation, to name a few.

Any emerging technology tends to follow the rule ascribed to increasing system complexity sometimes stated as 'more is different' with reference to the 1972 paper by Physicist Philip W. Anderson [?]. This especially applies to that of quantum networks. As greater quantum control is achieved, greater quantum networking capabilities will be required, thereby necessitating the development of improved methods for transmitting quantum information.

Currently, a hurdle in developing quantum networks is transmitting quantum information with a sufficiently long coherence time such that the information produced may actually be used. This is especially true if the information is to be transmitted across vast distances through optical fibres, such as China has done in a 4600 kilometer land-based network [?].

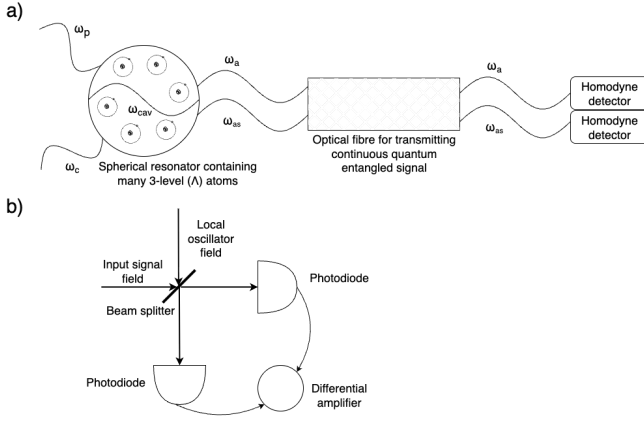


FIG. 1: Suggested setup of a real-world implementation for the four wave mixing process described with (a) two continuous beams entangled beams (continuous-variable entanglement) at frequencies ω_p and ω_c which undergo a four wave mixing process within a spherical resonator tuned to frequency ω_{cav} filled with cooled three-level atoms to produce Stokes and Anti-Stokes beams at frequencies ω_a and ω_{as} , respectively. The resulting beams are then may then be sent through an optical fibre to homodyne detectors. (b) Homodyne detector layout.

One dimension of this problem is that even if a coherent optical field is produced, it may not be at a frequency for which optical fibres have a low-loss window. Thus, effective frequency conversion methods which preserve quantum information must be developed.

This thesis includes a theoretically viable method for frequency conversion of two entangled electromagnetic fields (continuous-variable entanglement). The key idea is to send the entangled beams into an optical spherical resonator filled with cooled three-level atoms (Λ -system) such that a four wave mixing (FWM) process occurs to produce two entangled output fields. If the output frequencies are tuned such that they incur low-loss within an optical fibre, then the signals could theoretically be transmitted over long distances with existing technology. For a diagrammatic representation of this process, see figure 1. **TODO - Should squeezed states be mentioned here, or is it fine to leave them for later?**

The outline of this project can be described in three stages: formulating up the system Hamiltonian, computationally solving for energy eigenstates of the Hamiltonian, and analyzing the results to find how changing specific parameters may change the system behavior.

III. BACKGROUND INFORMATION

A. Electromagnetically Induced Transparency

In order to gain an intuition on electromagnetically induced transparency (EIT), we will start with a classical toy model of the atom. As is a common theme in physics, a hydrogenic atom in a time dependent electromagnetic field can be approximately modelled as a driven harmonic oscillator. So rather than dealing with quantized photons, we will consider a classical oscillating electromagnetic fields with frequencies nearly resonant with those of the atomic transitions.

To do this, a few assumptions will be made [?]: first, the wavelengths of the light sources interacting with the atom have wavelengths much greater than the Bohr radius of the atom ($\lambda \gg a_0$); second, the detunings of incoming beams are small in proportion to the transition frequencies considered. These two assumption happen to be consistent with one another.

The first assumption allows for us to approximate the electromagnetic field experienced by the atom as constant in space but varying in time since the change in electromagnetic field strength between the two ends the atom in the transverse wave direction is relatively small.

There exists the relation between the electric and magnetic field strengths of light which arises from Maxwell's equation [?]

$$B = \frac{E}{c}. \quad (1)$$

The Lorentz force is

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \quad (2)$$

where q is charge, and $\mathbf{v}$ is particle velocity. Using the Bohr model for approximation, we have that quantized electron velocities for a hydrogenic atom have magnitude

$$v_n = \alpha \frac{Z}{n} c \quad (3)$$

where α is the fine structure constant, Z is the atomic number, and n is the principle quantum number [?]. We can therefore see by rough approximation that we have

$$|\mathbf{v} \times \mathbf{B}| \lesssim \alpha Z c \frac{E}{c} = \alpha Z E. \quad (4)$$

We are considering Rubidium with atomic number $Z = 37$, but this neglects the effects of screening caused by the electrons in inner shells of the atom. Electron position is distributed in a cloud about the atom for any orbital, though we can expect the 36 electrons of inner shells between the valence electron (singular in its shell) and the nuclear charge of $37e$ to roughly cancel $36e$ of charge such that the valence electron experiences about $1e$. By this line of reasoning we have

$$|\mathbf{v} \times \mathbf{B}| \lesssim \alpha E. \quad (5)$$

Thus, the force due to the external magnetic field is about 1/137 that of the external electric field. So we may neglect the external magnetic field for simplification.

$$\mathbf{F} \approx q\mathbf{E} \quad (6)$$

All of these approximations can be justified from the perspective that this model is a first estimate for system behavior which will be refined and iterated upon.

This brings us to a physicist's favorite abstraction: the harmonic oscillator. The potential of which the valence electron experiences can be expanded in a Taylor series and approximated as quadratic [?]. The mass of the nucleus is much greater than the electron mass, so we may approximate the nucleus to be fixed. So we have a mass with some equilibrium distance from a fixed point under a quadratic potential, i.e. a simple harmonic oscillator. We may then add in our external time-dependent driving force to obtain a driven harmonic oscillator.

When a charge oscillates, it emits photons as its movement produces a changing electric and magnetic field. So when our system is driven on resonance, then an electromagnetic field of that same frequency will be produced. Here lies the idea behind electromagnetically induced transparency. By driving a cloud of these hydrogenic atoms on resonance, the field essentially 'passes through' the medium.

TODO - Talk about slowed light (group velocity) due to EIT in medium

B. Structure of atomic spectra

Atomic states for the electron of a hydrogenic atom are enumerated by the quantum numbers n (principle quantum number), l (angular momentum quantum number), m_l (magnetic quantum number), and m_s (spin quantum number) [?].

$$n \in \mathbb{N}, \quad l \in \{0, 1, \dots, n-1\}, \quad m \in \{-l, \dots, l\} \quad (7)$$

The energy levels for an unperturbed hydrogenic atom modelled with just the coulomb potential are quite simple as they just depend on the principal quantum number.

$$E_n = -\frac{(Z\alpha)^2 mc^2}{2n^2} = -\frac{Z^2 Ry}{n^2} \quad (8)$$

where Z is the atomic number, α is the fine structure constant, m is the electron mass, and c is the speed of light [?]. These unperturbed energy levels each have degeneracy of n^2 .

The real picture, is more complicated, however. There are fine structure and hyperfine structure perturbations on these energy levels, splitting energy levels and decreasing their degeneracies.

1. Fine Structure Corrections

Fine structure corrections arise due to relativistic corrections to the non-relativistic Schrödinger's equation. In particular, we have a pure relativistic correction from considering electron momentum (which is a non-negligible fraction of the speed of light), a spin-orbit coupling correction, and the Darwin correction. To first order perturbation theory, the combined energy shift of these contributions is

$$\Delta E_{fs} = \frac{(Z\alpha)^4 mc^2}{2} \frac{1}{n^3} \left(\frac{3}{4n} - \frac{2}{2j+1} \right) \quad (9)$$

where $j = |l \pm \frac{1}{2}|$ [?]. Notice that these perturbations are scaled by a factor of $(Z\alpha)^2$ in comparison to Eq. (8) such that for atoms with a sufficiently large atomic number, these approximate energy shifts arrived at by perturbation theory break down.

2. Hyperfine Structure Corrections

There exist yet smaller energy shifts due to the interaction between the internal magnetic fields of the nucleus and the orbital motion of the electron along with its spin [?]. For our purposes, it is sufficient to consider only hyperfine splitting for $l = 0$ states, which have the correction term

$$\Delta E_{hfs} = \frac{2}{3} \frac{m}{M} (Z\alpha)^4 \frac{mc^2}{n^3} [F(F+1) - s(s+1) - I(I+1)] \quad (10)$$

where I is the nuclear spin quantum number, s is the electron spin quantum number, and F is the total angular momentum quantum number with operator $F = \hat{I} + \hat{S}$ [?].

C. Atomic transitions

In a hydrogenic atom, the transitions between states are mediated by excitations in the electromagnetic field (i.e. photons). It is crucial in atomic physics to understand the probabilities of these atomic state transitions.

The interaction between an atom and the electromagnetic field is not a linear interaction, creating difficulty in finding an exact solution. To obtain approximate behavior, first order perturbation theory can be applied to dominating effects of this interaction [?]. From this, one obtains the transition dipole matrix with elements

$$\mu_{j,i} = \langle j | \hat{\mathbf{d}} | i \rangle = e \langle j | \hat{\mathbf{r}} | i \rangle \quad (11)$$

where $\hat{\mathbf{d}} = e\hat{\mathbf{r}}$ is the dipole transition operator and $\hat{\mathbf{r}}$ is the position operator [?].

If $\mu_{j,i} = 0$, then the transition between states $|i\rangle$ and $|j\rangle$ is a ‘forbidden’ dipole transition, otherwise the transition is an allowed dipole transition. This criterion enforces the dipole selection rules that

$$\Delta l = \pm 1, \Delta m_l = 0, \pm 1. \quad (12)$$

It is important to note that these transition rules are not always perfectly enforced. Since perturbation theory was applied to obtain them, there exist transitions which disobey these rules, however, these transitions occur with significantly decreased probability.

D. Three-Level Atom

In order to meaningfully utilize the power of quantum information via quantum states, it is beneficial to work with simplified, yet somewhat abstract, systems. It would be ideal if we could realize any imaginary quantum systems that exactly match our mathematical descriptions. Reality is not so convenient, however. We must work within the constraints enforced by the available quantum systems in terms of allowed states and transitions between those states.

With enough ingenuity, we can use existing systems to approximate useful mathematical abstractions. Enter, the three-level (hydrogenic) atom. All hydrogenic atoms, in theory, have infinitely many states up to ionization ($n \in \mathbb{N}$), and transition probabilities between many of these states can be non-zero. Though by finding a subset of states with desired characteristics and leveraging transition rules where any undesired state transitions are suppressed, a physical atom can be used to approximate a model system. This becomes especially useful in the limit of a large number of atoms, where statistically the desired transitions are the most probable, and any undesired transitions are vastly outnumbered, giving approximately the desired behavior.

A three-level atom has three distinct states with some set of allowed transitions between those states. There exist Λ -systems (see Fig. 2), V-systems (see Fig. 3), and Ξ -systems (see Fig. 4). From this point on we will focus on Λ -systems, which can be created with Rubidium TODO - Which Rb? How is this kind of system formed? [?].

The transition amplitude between states $|1\rangle$ and $|2\rangle$ is suppressed, so if the system is in state $|2\rangle$, then in order to transition back to state $|1\rangle$ the system will likely need to transition through the excited state $|3\rangle$. This makes $|2\rangle$ a quasi-stable state.

If there were a way to suppress all transitions to state $|3\rangle$, then we would obtain a system in which the atom is bound to states $|1\rangle$ and $|2\rangle$ such that any photons which would stimulate a transition to the excited state cannot be absorbed.

There is, in fact, a way to do just this. By driving the system with external electromagnetic fields (lasers)

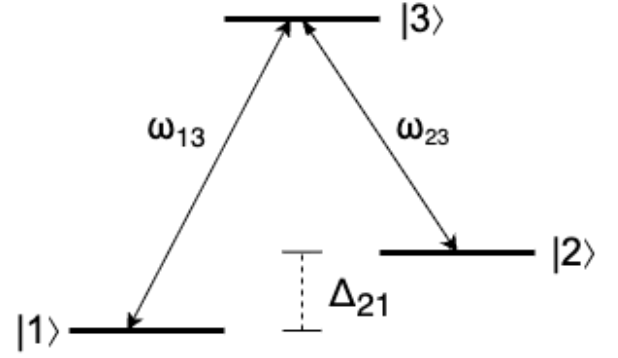


FIG. 2: TODO

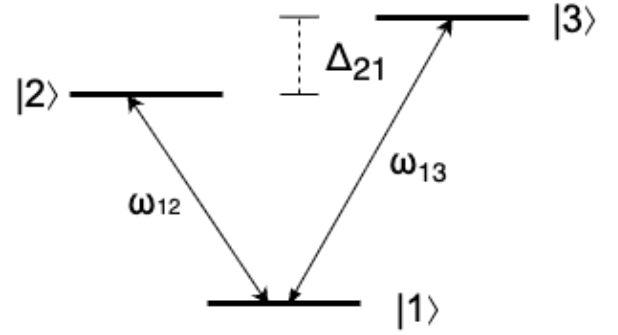


FIG. 3: TODO

at the transition frequencies, destructive quantum interference between the two fields along with specific initial conditions results in zero probability for the system to be in the excited state [?]. $|3\rangle$ is then called a ‘dark state’. In practice, since perfect precision in real-world systems is difficult, both lasers are detuned from the exact transition frequency. The probe laser stimulates transition 13 $\omega_p = \omega_{13} + \Delta$, $\Delta \ll \omega_{13}$, and the control laser stimulates

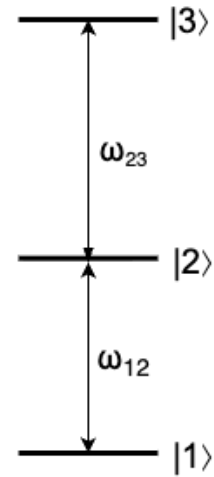


FIG. 4: TODO

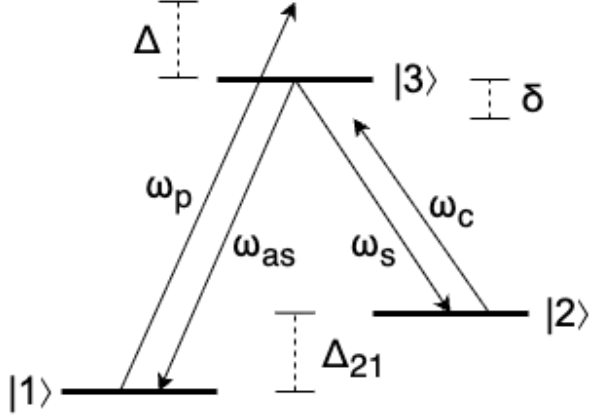


FIG. 5: TODO

transition 23 $\omega_c = \omega_{23} + \delta$, $\delta \ll \omega_{23}$ (see Fig. 5).

The main idea is that the system then becomes effectively ‘transparent’ to the probe and control laser frequencies. This is how EIT can be produced in practice.

To understand how this would be useful in practice, suppose you have a cloud of rubidium atoms at some temperature T (high enough such that they are in a gaseous state, say around 300K to 400K) in the absence of any external electromagnetic fields (i.e. no Stark or Zeeman effects), then by the principles of statistical mechanics, the ground state is the most common state. So the majority of atoms will sit in one of the two long-lived states ($|1\rangle$ or $|2\rangle$), while a smaller proportion of atoms are in the excited state $|3\rangle$. If the probe and control lasers are then powered on, we should get an approximate dark state where the majority of atoms are in states $|1\rangle$ or $|3\rangle$ while hardly any sit in $|2\rangle$. There may exist some atoms in higher excited states, but at a negligible population. This cloud of atoms will then exhibit EIT.

E. Cavity enhanced EIT

Consider a system composed of Avogadro’s number of three-level atoms sitting. In the ideal scenario, all atoms would have the exact same transition frequencies such that when the probe and control beams are fired through the cloud, the frequency of transparency is exact. In reality, this is not so. The system needs to be in a gaseous excited state where the electromagnetic fields of nearby atoms interact, inducing Stark and Zeeman shifts in their spectra, and Doppler shifts in the observed frequency of light would occur as some atoms would be moving away from the laser while some would be moving towards it. In addition, the laser used will have some bandwidth of emission frequencies. These effects combine to cause spectral broadening in the transparency window allowed by EIT.

Before moving on, we need an important fact. When light travels through some material with complex electric

susceptibility (i.e. complex index of refraction), then the light will receive some phase shift due to the real-valued component of susceptibility χ_R and some decay due to the imaginary-valued component χ_I , such that the oscillating electric field will propagate as [?]

$$E = E_0 e^{i(kz - \omega t)} e^{i\chi_R kz/2} e^{-\chi_I kz/2} \quad (13)$$

where the rate of decay then grows exponentially with distance traveled z . The decay term $e^{-\chi_I kz/2}$ is known as Beer-Lambert law [?].

Consider placing the atomic gas in an interferometric cavity (such as a Fabry-Perot cavity), the transparency window can be greatly narrowed. This can be thought of classically or quantum mechanically: in a classical interpretation, Beer’s law (Eq. (13)) will cause any frequencies which are slightly off resonant will decay after many reflections within the cavity (the path length is effectively longer); in a quantum interpretation, non-resonant states of light will destructively interfere within the cavity, leading to a decreased probability that they propagate.

The following are meeting notes

Imagine a semi-classical EIT medium (what we have) in a Fabry-Perot cavity.

Look at optics notes for Fabry-Perot cavity math.

What if instead we had the light pass through a medium of atoms? We end up with a phase shift from the atoms in the cavity and an attenuation unless you are perfectly resonant.

Fabry-Perot cavity gives a phase shift of e^{ikL_F} while atoms give a phase shift of e^{inkL_A} . Multiply the classical Fabry-Perot equation by e^{inkL_A} . $E_0 \approx te^{ikL_F} e^{inkL_A}$

We don’t need the Hamiltonian for the Fabry-Perot cavity. QM gives us n , so plug that value into the cavity equation. Phase shift should be due to Fabry-Perot and Beer’s Law. The strength inside the cavity will be very large. The strength into and out of the cavity will be about equal.

F. Optical Nonlinearities

The polarization of a material is often written as a linear expression with electric field strength

$$P = \epsilon_0 \chi E \quad (14)$$

but for a sufficiently large electric field, there is a saturation in the polarization strength [?]. We can account for this by expanding the expression as a series of the form

$$P = \epsilon_0 (\chi^{(1)} E + \chi^{(2)} E^2 + \chi^{(3)} E^3 + \dots) \quad (15)$$

where the i in $\chi^{(i)}$ is a superscript denoting the power to which the susceptibility term is associated where we would then call χ^i the i th order nonlinear optical susceptibility [?].

G. Four Wave Mixing

H. Squeezed States

According to Heisenberg's uncertainty principle, the product of the standard deviations of two canonically conjugate observables, x and p , must satisfy

$$\Delta x \Delta p \geq \frac{\hbar}{2}. \quad (16)$$

For light of sufficient intensity, one can consider its amplitude and phase as canonically conjugate variables [?]. For a generically produced beam, one would expect the standard deviations of amplitude and phase to be roughly equal. If measurement were to be performed in a single quadrature (amplitude for example), then reducing the uncertainty in that quadrature would result in a more precise overall reading.

This leads to the idea of a squeezed state in which one quadrature has its uncertainty decreased below the shot noise level at the expense of the other, such that the uncertainty principle is still observed. It has been shown that optical squeezed states can be produced through a FWM process in optical cavities [?].

I. Continuous Variable Quantum Information

If multiple beams are entangled such that a measurement of one of these variables on one beam yields information about the other beams, then we have what is called continuous variable entanglement [?]. In this way, quantum information can be stored, processed, and transmitted using continuous beams rather than with individual photons.

J. How Sub-Poissonian Light and Photon Anti-Bunching Effects Affect Continuous Variable Measurements

Continuous variable measurements allow for small quantum fluctuations to be less significant in measurement procedures when compared to individual photon entanglement, though there still exist challenges in performing measurements. Continuous variable measurements are typically performed with homodyne detectors [?], a type of interferometer which compares some input beam against a controlled probe beam and takes the difference between their readings [?].

Light is quantized and beam intensity is not that high. Individual photons arrive at the detector in a Poisson process, resulting in shot noise for the detector [?]. Due to this, the reading of the input beam can be swayed by individual photon arrival events. It would be highly desirable for the input beam to follow sub-Poissonian statistics (with the same mean as a Poisson distribution but a

lower variance) [?] and for the beam to display photon anti-bunching (arrival events are distributed more evenly as opposed to having intensity spikes) [?].

It has been shown that applying cavity EIT to three-level atoms can produce sub-Poissonian light [?], and that applying FWM to three-level systems can produce photon anti-bunching effects [?]. Combining these techniques theoretically allows for the information of continuous variable beams to be more reliably transmitted to some measurement device, such that readings fall below the shot-noise level.

K. Fock Basis

L. Master Equation

- Born-Markov approximation (Born = weak interactions between bath and system. Markov = Memoryless system)
- Applying rotating wave approximation to Born-Markov equation for $\dot{\rho}(t)_S$ yields Lindblad Master equation
- Dissipative operators

M. QuTiP Python Package

IV. THE FIRST STEPS

In order to work up to modelling a three-level atom in the Fock basis for photons with dissipative Lindbladian operators, we will start with a simpler model, then iterate upon it towards increasing complexity. We will treat our atom as a three-level system with discrete states while light behaves as a continuous field such that we do not need to discretize light in the Fock basis of photon number.

The polarization vector of the atomic cloud may be given by

$$\mathbf{P} = N \langle \hat{\mathbf{d}} \rangle = \epsilon_0 \chi \mathbf{E} \quad (17)$$

where N is atomic density, $\hat{\mathbf{d}}$ is the average dipole moment, and χ is the magnetic susceptibility [?]. One can find that for a three-level atom (TODO derive or cite?)

$$\chi = \frac{-Nd^2}{\epsilon_0 \hbar} \frac{\delta}{|\Omega_c|^2 - \delta(\Delta + i\Gamma)} \quad (18)$$

where TODO - Define detunings and dissipation rate along with a figure for a three-level atom.

The Hamiltonian of the three-level atom can be given as

$$H_{atom} = TODO(diagonal - slides) \quad (19)$$

and the Hamiltonian of the light-matter interaction can be given as

$$H_{atom} = \text{TODO}(\text{see slides}) \quad (20)$$

The dissipative operators...

Solve master equation with...

Here's the plot...

A. Meeting 2 with MacRae

1. Hamiltonian Derivation with light as a classical field

To construct the Hamiltonian, it is advantageous to first express it as a sum of the contributions to be considered

$$\hat{H} = \hat{H}_{Atom} + \hat{H}_{Light\ Atom\ Interaction} + \hat{H}_{Light}. \quad (21)$$

Since we are considering the simplified system where light acts as a continuous field, all we care about is how that field interacts with the atom. So at this point, we choose to neglect the term $\hat{H}_{Light}$ to obtain

$$\hat{H} = \hat{H}_{Atom} + \hat{H}_{Light\ Atom\ Interaction}. \quad (22)$$

Let $\hat{\sigma}_{i,j} = |i\rangle\langle j|$ express the atomic transition operators which operate on state vectors to transition from state j to i . For example,

$$\hat{\sigma}_{1,3} = |1\rangle\langle 3| = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad (23)$$

and acting this operator on basis state $|i\rangle$ yields

$$\hat{\sigma}_{1,3}|i\rangle = |1\rangle\langle 3|i\rangle = \delta_{3,i}|1\rangle \quad (24)$$

where δ denotes the usual Kronecker delta.

TODO

3-level system frequency $2-3 = \Omega_c$. $\Gamma_{3 \rightarrow 2}$ and $\Gamma_{3 \rightarrow 1}$ emission. $\hat{\sigma}_{i,j} = |i\rangle\langle j|$. $\Delta\omega_s - \omega_{3,1}$. $\delta = \Delta_c - \Delta$

$$\hat{H} = \hat{H}_a + \hat{H}_{LightAtom} + \hat{H}_{Light}$$

Neglect $\hat{H}_{Light}$

$$\hat{H}_a = \hbar(\omega_{3,1} - \omega_{3,1})\hat{\sigma}_{2,2} + \hbar\omega_{3,1}\hat{\sigma}_{3,3}$$

$$\hat{H}_{LightAtom} = -\hat{\mathbf{d}}_{1,3} \cdot \mathbf{E}_s - \hat{\mathbf{d}}_{2,3} \cdot \mathbf{E}_c$$

Expand dipole operators and E in terms of complex exponentials.

$$\hbar\Omega_s = \mathbf{d}_{1,3} \cdot \mathbf{E}_s$$

$$\hbar\Omega_c = \mathbf{d}_{2,3} \cdot \mathbf{E}_c$$

Plug in Ω_i values into hamiltonian, replacing dipoles and E

We need the interacting picture

$$\hat{H} = \hat{H}_0 + \hat{H}_I$$

$$\hat{H}_0 = \omega_s\hat{\sigma}_{3,3} + (\omega_s - \omega_c)\hat{\sigma}_{2,2}$$

$$\frac{\hat{H}_I}{\hbar} = \frac{\hat{H} - \hat{H}_0}{\hbar} = (\omega_{3,1} - \omega_s - \omega_s + \omega_{3,2})\hat{\sigma}_{2,2} + (\omega_{3,1} - \omega_s)\hat{\sigma}_{3,3} + \hat{H}_{LightAtom}$$

Plug in Δ and δ then $\hat{H}_{LightAtom}$ to obtain interaction hamiltonian

$$\hat{H}_I = \begin{pmatrix} 0 & 0 & \Omega_s e^{i\omega_s t} \tilde{\Omega}_s e^{i\omega_s t} \\ 0 & \delta & \Omega_c e^{i\omega_c t} \tilde{\Omega}_c e^{i\omega_c t} \\ c.c & c.c & \Delta \end{pmatrix}$$

Then

$$V = e^{-i\hat{H}_0 t} \hat{H}_I e^{i\hat{H}_0 t}$$

Notice that $e^{-i\hat{H}_0 t}$ is diagonal so this is simple

Multiply through, then apply rotating wave approximation to obtain

$$\hat{V} = \begin{pmatrix} 0 & 0 & \Omega_s \\ 0 & \delta & \Omega_c \\ c.c & c.c & \Delta \end{pmatrix}$$

2. a

Density matrix times an operator

$$\hat{\rho}\hat{A} = |\psi\rangle\langle\psi|\hat{A}$$

$$\hat{B} = I\hat{B} = \sum_c |c\rangle\langle\hat{B}|c\rangle^\dagger$$

Do some math, then we find that

$$\langle\hat{A}\rangle = \text{Tr}[\hat{\rho}\hat{A}]$$

For linear media, the polarization is $\mathbf{D} = \epsilon_0\mathbf{E} + \mathbf{P} = \epsilon_0(1 + \xi)\mathbf{E} = \epsilon\mathbf{E}$ and $\mathbf{P} = N\langle\hat{\mathbf{d}}\rangle$

$$1 + \xi = \frac{\epsilon}{\epsilon_0}$$

For non-magnetic materials $\mu \approx \mu_0$, so

$$1 + \xi = \frac{\epsilon}{\epsilon_0} = \frac{\epsilon\mu}{\epsilon_0\mu_0} = \left(\frac{c}{v}\right)^2 = n^2$$

Something like this, check all these derivations

Write up derivation to the point that we obtain

$$\xi = \frac{N\langle\hat{\mathbf{d}}\rangle}{\epsilon_0\hbar\Omega}$$

$$\langle\mathbf{d}\rangle = \text{Tr}[\hat{\rho}(\hat{\mathbf{d}}\hat{\sigma}_{i,j} + h.c.)]$$

Expand out (Find a textbook with this) and we obtain

$$\xi = \frac{Nd^2}{\epsilon_0\hbar\Omega}\rho_{i,j}$$

The density matrix should be a unitary

B. Meeting 3 with MacRae

Note that Polarization is real as $P = P_0 e^{i\omega t} + P_0^* e^{-i\omega t}$ as real valued

We use the result we obtained $\langle d \rangle$ with Beer's Law to obtain $E(t)$ (wave propagation, phase shift, and absorption). We get no phase shift and no absorption for perfect EIT. In reality, we send a pulse through. Some of the pulse frequencies will be absorbed! The longer the pulse, the less we lose. This is the effect of the medium responding.

For a pulse of length $\Delta\tau$. If $\Delta\tau = 10^{-5}$ then the light pulse will be about 3 km long! We need the pulse to fit in the EIT window. Slowed light helps here such that the pulse fits in the cell. Widening the frequency window increases the speed of the slowed light, so we need a very precise frequency. The narrower the transparency

window, the greater the group velocity (see the equation for group velocity dependent on $dn/d\omega$).

We can however include many more atoms to increase n to lower group velocity. This is called ‘increasing optical depth’.

$$v_g = \frac{c}{n + \omega \frac{dn}{d\omega}} \quad (25)$$

V. TEMP SECTION - NEXT STEPS

I have covered a sufficient amount of background and written enough on it. My meeting notes document to sufficient detail that I can write them out formally to approximately the same degree now vs later. If anything, they will benefit from being written later with hindsight into how things actually work.

So, I can leave the thesis here for now. My next steps then are the following:

- Read and re-read relevant CEIT papers
- Find how to model the three-level system within a cavity using classical electromagnetic fields with dissipative operators for Lindbladian time evolution
- Find how to quantize the system (optional - but probably in the papers I will read)
- Read FWM papers
- Learn how to computationally model FWM

VI. SEMI-CLASSICAL THREE-LEVEL ATOM

We can begin to understand the behavior of EIT in three-level Λ -type atoms through semi-classical modelling. We can consider a single three-level atom with quantized states in the presence of a classically oscillating electromagnetic field

$$\vec{E} = \vec{E}_0 e^{i\omega t}. \quad (26)$$

In this limit, we consider E_0 to be sufficiently large such that photon quantization can be neglected (i.e. in the classical limit, the behavior of the quantized electromagnetic field must converge to the classical behavior). We may express the full Hamiltonian as the sum

$$\hat{H} = \hat{H}_{Atom} + \hat{H}_{Light} + \hat{H}_{Light\ Atom\ Interaction}. \quad (27)$$

We may neglect the term $\hat{H}_{Light}$ to approximate our full Hamiltonian since the state of the electromagnetic field is not significantly affected by the atom in the (i.e. there is little coupling between the state of the

laser field and the single atom). Our Hamiltonian then becomes

$$\hat{H} = \hat{H}_{Atom} + \hat{H}_{Light\ Atom\ Interaction}. \quad (28)$$

where **TODO - Define ω values?**

$$\begin{aligned} \hat{H}_{atom} &= \omega_s \hat{\sigma}_{3,3} + (\omega_s - \omega_c) \hat{\sigma}_{2,2} \\ &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & \omega_s - \omega_c & 0 \\ 0 & 0 & \omega_s \end{pmatrix} \end{aligned} \quad (29)$$

and

$$\hat{H}_{Light\ Atom\ Interaction} = -\hat{d}_{1,3} \cdot \vec{E}_s - \hat{d}_{2,3} \cdot \vec{E}_c \quad (30)$$

TODO - Finish rest of derivation!!

Thus, we obtain

$$\hat{H} = \begin{pmatrix} 0 & 0 & \Omega_s \\ 0 & \delta & \Omega_c \\ \Omega_s^* & \Omega_c^* & \Delta \end{pmatrix} \quad (31)$$

TODO - This has been coded, include code snippets here and talk about results.

VII. JANYES-CUMMINGS MODEL

Before moving on, it is useful to look at another simplified model, though one with very different characteristics and description. We may then combine the ideas from these models to form more complex systems.

Introducing the Jaynes-Cummings model. For this, we take a step back to consider a two-level atom, but add complexity by placing the atom within a resonant cavity (see Fig. 6) [?]. The field of a single photon within the cavity is quite significant as it will bounce back and forth between cavity walls, building up a non-negligible field. So we must quantize photons in the Fock basis. The cavity is finely tuned to a resonant frequency ω_c such that any other frequency of light destructively interferes with itself. This leads to a simplification for our Fock basis which may be restricted to photons of a single frequency. There can be a detuning between the frequencies, but we will consider the case of perfect resonance here $\omega = \omega_a = \omega_c$.

The states of our system live within the combined Hilbert space $\mathcal{H}_{atom} \otimes \mathcal{H}_{photon}$, and so states are described by $|atom\rangle \otimes |photons\rangle$ where $|atom\rangle$ has two states and $|photon\rangle$ describes all n -photon states. Consider a state containing N photons worth of energy. Then the possible states of the system are $|\downarrow\rangle \otimes |N\rangle$ and $|\uparrow\rangle \otimes |N-1\rangle$.

The Hamiltonian describing this system is given by [?] [?]

$$\hat{H}_{JC} = \hat{H}_{Atom} + \hat{H}_{Photon} + \hat{H}_{Interaction}. \quad (32)$$

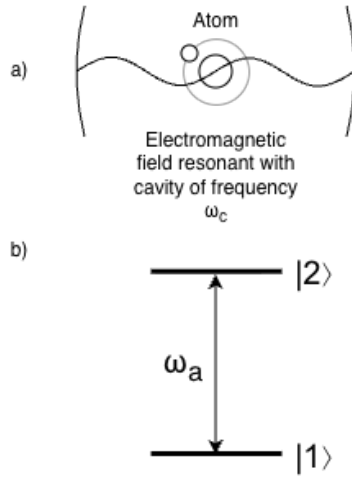


FIG. 6: Depiction of the system for a Jaynes-Cummings model with (a) a resonant cavity of frequency ω_c containing (b) a two-level atom of transition frequency ω_a .

The atomic term is given by

$$\begin{aligned}\hat{H}_{Atom} &= \frac{\hbar\omega}{2} \hat{\sigma}_z \\ &= \frac{\hbar\omega}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.\end{aligned}\quad (33)$$

The photonic term is given by

$$\hat{H}_{Photon} = \hbar\omega \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right).$$

where we may shift by the zero-point energy since it is a constant term to obtain

$$\hat{H}_{Photon} = \hbar\omega \hat{a}^\dagger \hat{a}$$

Then finally, the interaction term is given by

$$\begin{aligned}\hat{H}_{Interaction} &= \frac{\hbar\Omega}{2} (\hat{\sigma}_+ \otimes \hat{a} + \hat{\sigma}_- \otimes \hat{a}^\dagger) \\ &= \frac{\hbar\Omega}{2} \begin{pmatrix} 0 & \hat{a} \\ \hat{a}^\dagger & 0 \end{pmatrix}\end{aligned}\quad (34)$$

where Ω is the Rabi frequency and $\hat{\sigma}_\pm$ are the atomic raising and lowering operators.

In order to combine the Hilbert spaces for these two separate quantized systems, block-operator notation is applied. Notice that $\hat{H}_{Atom}$ acts on the two-state atomic basis, while $\hat{H}_{Photon}$ acts on the infinite Fock basis of photons. We then must combine the two bases in $\hat{H}_{Interaction}$ which describes the creation of a photon in the Fock basis upon atomic emission, and the annihilation of a photon upon absorption. So this Hamiltonian should not be thought of as a two-by-two matrix acting on the atomic basis, but rather an infinite matrix enumerating the interactions between all possible states. All together, the final Hamiltonian is then [?]

$$\begin{aligned}\hat{H}_{JC} &= \frac{\hbar\omega}{2} \hat{\sigma}_z + \hbar\omega \hat{a}^\dagger \hat{a} + \frac{\hbar\Omega}{2} (\hat{\sigma}_+ \otimes \hat{a} + \hat{\sigma}_- \otimes \hat{a}^\dagger) \\ &= \frac{\hbar}{2} \begin{pmatrix} \omega & \Omega \hat{a} \\ \Omega \hat{a}^\dagger & -\omega \end{pmatrix} + \hbar\omega \hat{a}^\dagger \hat{a}\end{aligned}\quad (35)$$

This is not the whole story, however. Any real cavity has some photonic decay rate, meaning we must include dissipative operators and apply Lindbladian dynamics to solve for a steady state via the master equation. Our dissipative operator then becomes $L = \sqrt{\gamma} \hat{a}$ where γ gives the decay rate such that our master equation is [?]

$$\dot{\rho} = -i [\hat{H}_{JC}, \rho] + \gamma \left(\hat{a} \rho \hat{a}^\dagger - \frac{1}{2} \hat{a}^\dagger \hat{a} \rho - \frac{1}{2} \rho \hat{a}^\dagger \hat{a} \right). \quad (36)$$

To gain experience working with the QuTiP package, the Jaynes-Cummings model was coded (see Code. VII) based on an implementation by Robert Johansson [?]. This code differs slightly from the above description in that the atomic Hamiltonian basis is shifted such that it is in the rotating frame $\hat{H}_{Atom} = \hbar\omega |e\rangle\langle e|$.

Code Block 1: Python code for the Jaynes-Cummings Model

```
# Initial Arguments
w_a = 2.0
w_c = 2.0
W = 0.01
gamma = 0.005
xi = 0.0025
eta = 0.05
h = 1.0
N = 10 # Truncate the Fock basis to
        N=10 photons
times = np.linspace(0, 2500, 10000)

# Set up operators in the relevant
    basis
a = tensor(qeye(2), destroy(N)) # Acts
    on photon Fock basis with N states
    as lowering operator
s_minus = tensor(destroy(2), qeye(N))
    # Acts on atomic basis with 2
    states as lowering operator
s_atom = tensor(Qobj([[0, 0], [0,
    1]]), qeye(N)) # Acts on atomic
    basis with a shifted ground states

# Set up the Hamiltonian
H = (h * w_a) * s_atom + (h * w_c) *
    a.dag() * a + (h * W) *
    (s_minus.dag() * a + s_minus *
    a.dag())

# Set up dissipative operator for a
    lossy cavity
cavity_decay = np.sqrt(gamma) * a
```

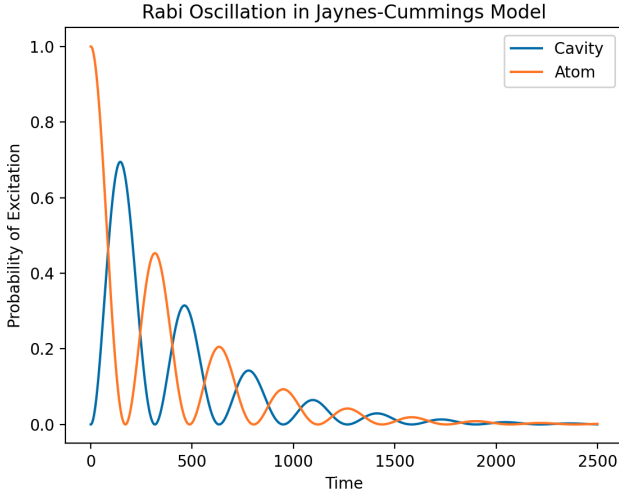


FIG. 7: Decaying Rabi oscillation produced by the Jaynes-Cummings model with initial state $|\psi\rangle_0 = |e\rangle \otimes |0\rangle$.

```
c_ops = [cavity_decay]

# Set up the expectation value operators
number_operator = a.dag() * a
atomic_state_operator = s_minus.dag() * s_minus
e_ops = [number_operator, atomic_state_operator]

# Start the system in an initially excited state with 0 photons
psi_0 = tensor(basis(2,1), basis(N, 0))

# Time evolve the system using qutip mesolve
result = mesolve(H, psi_0, times, c_ops=c_ops, e_ops=e_ops)
photons = result.expect[0]
```

For an initial state where the atom is excited and there are no photons present in the cavity, we obtain decaying Rabi oscillations as expected (see Fig. 7).

A. Cavity EIT with a Λ -System

Place in appendix

Code Block 2: Python code for CEIT with a three-level atom

```
# Initial Arguments
kappa = 1.0 # Write all parameters in terms of kappa
W_c = 40.0 * kappa
g = 50.0 * kappa
```

```
Delta = 0.01
epsilon = 1.0 * kappa
Gamma_31 = 0.1 * kappa
Gamma_32 = 0.1 * kappa
N = 10 # Truncate the Fock basis to N=10 photons

# Atomic basis kets
atomicKet1 = basis(3, 0) # |1>
atomicKet2 = basis(3, 1) # |2>
atomicKet3 = basis(3, 2) # |3>

# Set up operators in the relevant basis
s11 = tensor(atomicKet1 * atomicKet1.dag(), qeye(N))
s22 = tensor(atomicKet2 * atomicKet2.dag(), qeye(N))
s33 = tensor(atomicKet3 * atomicKet3.dag(), qeye(N))
s13 = tensor(atomicKet1 * atomicKet3.dag(), qeye(N)) # Photon absorption 1->3
s23 = tensor(atomicKet2 * atomicKet3.dag(), qeye(N)) # Photon absorption 2->3
s31 = s13.dag() # Photon emission 3->1
s32 = s23.dag() # Photon emission 3->2
a = tensor(qeye(3), destroy(N)) # Acts on photon Fock basis as lowering operator

# Decay operators
cavity_decay = np.sqrt(2*kappa) * a
atomic_decay_13 = np.sqrt(2*Gamma_31) * s13
atomic_decay_23 = np.sqrt(2*Gamma_32) * s23
c_ops = [cavity_decay, atomic_decay_13, atomic_decay_23]

def eta(n):
    return g * np.sqrt(n/2) / W_c

def steady_evo():
    state_1_vals = []
    state_2_vals = []
    state_3_vals = []
    state_4_vals = []
    state_5_vals = []
    Delta_vals = np.linspace(-100.0, 100.0, 1000)

    # Useful basis vectors
    k_s2_n0 = tensor(atomicKet2, basis(N, 0))
    k_s3_n0 = tensor(atomicKet3, basis(N, 0))
```

```

k_s2_n1 = tensor(atomicKet2,
    basis(N, 1))
k_s3_n1 = tensor(atomicKet3,
    basis(N, 1))

for Delta in Delta_vals:
    # Get system for Delta value
    # Note that we assume perfect
    # cavity resonance (Delta_1,
    # Delta_2 = 0)
    H = -Delta * s11 + \
        Delta * a.dag() * a + \
        epsilon * (a + a.dag()) + \
        g * (a * s31 + a.dag() *
            s13) + \
        W_c * (s32 + s23)

    # Compute steady state solution
    rho = steadystate(H, c_ops,
        method='direct')

    # Find phase for n=0 and n=1
    expect_n0 = expect(k_s2_n0 *
        k_s3_n0.dag(), rho)
    phi_n0 = np.angle(expect_n0)
    expect_n1 = expect(k_s2_n1 *
        k_s3_n1.dag(), rho)
    phi_n1 = np.angle(expect_n1)

    # Define rotated plus minus
    # states in full Hilbert
    # space
    plus_n0 = (k_s2_n0 +
        np.exp(-1j*phi_n0)*k_s3_n0)
        / np.sqrt(2)
    minus_n0 = (k_s2_n0 -
        np.exp(-1j*phi_n0)*k_s3_n0)
        / np.sqrt(2)
    plus_n1 = (k_s2_n1 +
        np.exp(-1j*phi_n1)*k_s3_n1)
        / np.sqrt(2)
    minus_n1 = (k_s2_n1 -
        np.exp(-1j*phi_n1)*k_s3_n1)
        / np.sqrt(2)

    # psi^(0)_1
    psi_1 = tensor(atomicKet1,
        basis(N, 1)) - \
        eta(1) * plus_n0 + \
        minus_n0

    # psi^(0)_2
    psi_2 = tensor(atomicKet1,
        basis(N, 2)) - \
        eta(2) * plus_n1 + \

```

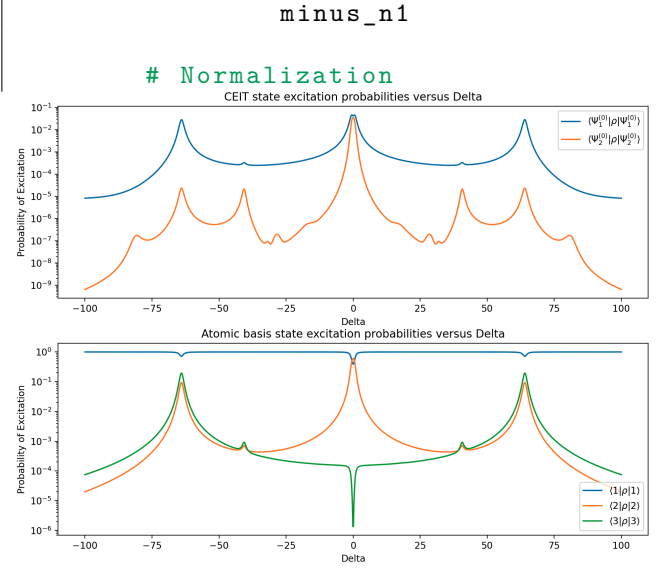


FIG. 8: TODO

```

psi_1 = psi_1.unit()
psi_2 = psi_2.unit()

# Operators
psi_1_op = psi_1 * psi_1.dag()
psi_2_op = psi_2 * psi_2.dag()

# Get expectation values
exp_1 = expect(psi_1_op, rho)
exp_2 = expect(psi_2_op, rho)
exp_3 = expect(s11, rho)
exp_4 = expect(s22, rho)
exp_5 = expect(s33, rho)
state_1_vals.append(exp_1)
state_2_vals.append(exp_2)
state_3_vals.append(exp_3)
state_4_vals.append(exp_4)
state_5_vals.append(exp_5)

```

VIII. HAMILTONIAN FORMULATION

IX. COMPUTATIONAL METHODS

X. RESULTS

XI. DISCUSSION

A. Conditions for Optical Squeezing